C_full internal scope closure (COMFORTABLE): B1/B2 Reading-H enacted as

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope: The enacted statement (comfortable)

B1/B2 Reading-H selection is enacted at tier T7 with the competitor class widened to the full admissible class $\mathcal{C}_{\text{full}}$, scope-qualified T7-SCOPE$_{\mathcal{C}_{\text{full}}}$:

$$F[Q] - F[G_*] > 0 \quad \forall\, Q \in \mathcal{C}_{\text{full}} \setminus \{G_*\}, \qquad (I, \mu^2) \in \mathcal{R}, \quad \text{given A1-KERNEL-CONV},$$

with $\mathcal{R} = \{0 < \mu^2 < \mu^2_{\max} = 0.0342,\ 0 < I < I_c^{\text{sel}}(\mu^2)\}$ the FULL Step-1 selection region. This is NOT an unrestricted T7 (§5).

2. Why the region is the full Step-1 region (the coherence upgrade)

The off-diagonal stability needs $K_{\text{floor}} < K_* = 20.55$. The coherence circle-packing lemma (this round) proves, elementarily and uniformly across $\mathcal{R}$,

$$K_{\text{floor}} \le T'(Q) \le \left\lfloor \frac{2\pi}{\theta_{\min}} \right\rfloor = 10 \quad \text{(every admissible competitor; } \rho \le Q_0,\ \text{pairwise } \ge \theta_{\min}\text{)},$$

so $R_{\text{lead}} \le 23.2 \cdot 11 \cdot I$: at $I = 2 \times 10^{-3}$, $R_{\text{lead}} \le 0.510$ ($\times 1.96$); at $I_c^{\text{sel}} = 2.41 \times 10^{-3}$, $R_{\text{lead}} \le 0.615 < 1$. The off-diagonal cap $I_{\text{off}}^{\text{coh}} = 1/(23.2 \cdot 11) = 3.92 \times 10^{-3}$ EXCEEDS I_c^{sel}, so the SELECTION boundary binds and $\mathcal{R}$ is the full Step-1 region with 20% headroom at the operating point. This SUPERSEDES the v1.1 thin margin $\times 1.026$ / cap 2.053×10^{-3} / 2.6% headroom, which were artefacts of the loose antipodal bound $T' \le N/2 \le 20$.

3. The three sectors at enactment (all comfortable)

sector	C_full status	margin
(D) diagonal isotropy	competitor-agnostic (Step 3)	comfortable ($\frac{1}{2} r_R^2,\ \frac{3}{2} u_{\text{eff}} > 0$)
(O) off-diagonal	coherence lemma $T' \le 10$ (this round)	**comfortable** ($R_{\text{lead}} \le 0.510$, $\times 1.96$)
(S) selection floor	competitor-agnostic (Step 3)	comfortable (joint 1.04, cap 1.13)
EXT	strong evidence	T2; **optional** (no longer needed for comfort)

All three sectors are now comfortable; the off-diagonal is no longer the thin ingredient. The sector caveats of v1.1 stand: (D) conditional on T-016 being a global/convex diagonal minimum; (S) on $J_{\text{eff}}, n_{\text{pack}}$ worst-case over the full class.

4. Gate status

G1 off-diagonal margin — **CLOSED by PROOF** (coherence lemma; comfortable $\times 1.96$, not merely accepted-thin).

G2 external referee / reproduction — OPEN, a validation/publication gate, not an internal proof blocker.

G3 operator sign-off — **CLOSED** (enacted T7-SCOPE$_{\mathcal{C}_{\mathrm{full}}}$).

EXT (T2) is now an OPTIONAL tightening ($\times 1.96 \to \times 5.4$), not required.

5. Permitted and forbidden claims

Permitted: "Within the certified operating region $\mathcal{R}$, Reading-H selection holds against all admissible $\mathcal{C}_{\mathrm{full}}$ competitors (given the A1 kernel convention), with a comfortable off-diagonal margin."
Forbidden: "Reading-H is globally / unconditionally proven over all intensities and all possible competitors." Still scope-qualified: within $\mathcal{R}$, given A1, external referee pending.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"Does the coherence lemma really hold uniformly?"* Yes — $2\pi/\theta_{\min} \leq 10.54$ across $\mathcal{R}$ (res5_036 scan), so $T' \leq 10$; the proof uses only $\rho \leq Q_0$ and pairwise $\geq \theta_{\min}$ (elementary).

(β) *"Is the comfortable upgrade an over-correction of the accepted thin margin?"* No — it is strictly stronger and rigorous: the thin $\times 1.026$ used the loose $T' \leq N/2$; the coherence $T' \leq 10$ is tighter and proved. The enactment is preserved, the margin improved.

(γ) *"Unrestricted T7?"* No — §5; scope-qualified within $\mathcal{R}$, given A1, referee pending.

Quantitative sanity check (mandatory): $T' \leq \lfloor 2\pi/\theta_{\min} \rfloor = 10$ (sup 10.54); adversarial max 8; $R_{\mathrm{lead}} \leq 0.510$ ($\times 1.96$), ≤ 0.615 at the selection cap; $I_{\mathrm{off}}^{\mathrm{coh}} = 3.92 \times 10^{-3} > I_c^{\mathrm{sel}} = 2.41 \times 10^{-3}$ (selection binds). All from res5_036 (5/5).

7. Result footer

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Result ID:                B1/B2 C_full internal scope closure
                         (COMFORTABLE) -- T7-SCOPE_{C_full}
Precise statement:        Coherence circle-packing lemma (res5_036 5/5):
                         T'<=floor(2pi/theta_min)=10
                         uniformly for every admissible competitor
                         (lattice + non-lattice) =>
                         K_floor<=10 => R_lead<=0.510<1 at I_op (x1.96),
                         0.615 at the selection cap
                         -- COMFORTABLE. I_off^coh=1/(23.2*11)=3.92e-3 >
                         I_c^sel=2.41e-3 => SELECTION
                         binds => C_full region = full Step-1 region R
                         (20% headroom). B1/B2 enacted
                         T7-SCOPE_{C_full} (thin-O qualifier removed).
                         (D)/(S) competitor-agnostic
                         comfortable. G1 CLOSED by PROOF, G3 CLOSED, G2
                         (referee) validation gate.
                         EXT T2 optional (no longer needed). Supersedes
                         v1.1 thin x1.026. Given A1.
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Scope:	T7-SCOPE_{C_full} within R, given A1; comfortable. External referee pending.
Dependencies:	res5_036 (coherence lemma), res5_032 (Step 1 R), res5_034 (Step 3 (D)/(S)), R-025 (K_floor<=T'); supersedes res5_033/035 thin numbers.
Evidence grade:	PROVED (elementary coherence lemma) + INTERNAL SCOPE CLOSURE + EXECUTED.
Reproduction command:	python codes/vacuum/res5_036_coherence_offdiag_bound.py
Expected output:	T'<=10; adversarial max 8; R_lead<=0.510 (x1.96); selection binds; 5/5 PASS.
Falsifier:	An admissible competitor with a sum-circle of >10 points; theta_min<0.571 in the window; or R_lead>=1 for some (I,mu^2) in R.
Tier before / after:	B1/B2 T7-SCOPE_{C_full, thin O} -> T7-SCOPE_{C_full} (comfortable). EXT T2 optional. NOT unrestricted T7.
No-overclaim statement:	Comfortable but scope-qualified (within R, given A1, C_full class). NOT global/all-intensity/all-competitor. EXT remains a separate optional T2.
Next required action:	G2 external/second-author referee (validation). EXT optional (further x5.4).